

# Proving Equations Using Identities

## The Basic Trigonometric Identities

### Reciprocal Identities

$$\sin \theta = \frac{1}{\csc \theta} \quad \csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta} \quad \sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta} \quad \cot \theta = \frac{1}{\tan \theta}$$

### Ratio Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

### Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

Prove that each equation is true by converting the left side to the right.

• ①  $\cos \alpha \sin \alpha \tan \alpha = \sin^2 \alpha$  ②  $\tan w \cot w \sec w = \frac{1}{\cos w}$

$$\cancel{\cos \alpha} \sin \alpha \frac{\sin \alpha}{\cancel{\cos \alpha}} =$$

$$\sin^2 \alpha =$$



$$\cancel{\tan w} \frac{1}{\cancel{\tan w}} \sec w =$$

$$\sec w =$$

$$\frac{1}{\cos w} =$$



③  $\cos^2 \beta + 5 + \sin^2 \beta = 6$  ④  $7 + \sec^2 x = 8 + \tan^2 x$

$$\sin^2 \beta + \cos^2 \beta + 5 =$$

$$1 + 5 =$$

$$6 =$$



$$7 + \tan^2 x + 1 =$$

$$8 + \tan^2 x =$$





• (5)  $\sin \theta \cot \theta \cos \theta = \cos^2 \theta$  (6)  $\csc A \tan A \sin A = \frac{1}{\cot A}$

$$\cancel{\sin \theta} \frac{\cos \theta}{\cancel{\sin \theta}} \cos \theta =$$

$$\cos^2 \theta =$$



$$\frac{1}{\cancel{\sin A}} \tan A \cancel{\sin A} =$$

$$\tan A =$$

$$\frac{1}{\cot A} =$$



(7)  $\sin^2 B - 2\sin B + \cos^2 B = 1 - 2\sin B$  (8)  $\csc^2 x + 2 = 3 + \cot^2 x$

$$\sin^2 B + \cos^2 B - 2\sin B =$$

$$1 - 2\sin B =$$



$$1 + \cot^2 x + 2 =$$

$$3 + \cot^2 x =$$



Using the Distributive Property

The Distributive Property:  $a(b+c) = ab+ac$

• (9)  $\cos \theta (\csc \theta + \tan \theta) = \cot \theta + \sin \theta$

$$\cos \theta (\csc \theta + \cos \theta \tan \theta) =$$

$$\cos \theta \frac{1}{\sin \theta} + \cos \theta \frac{\sin \theta}{\cos \theta} =$$

$$\frac{\cos \theta}{\sin \theta} + \sin \theta =$$

$$\cot \theta + \sin \theta =$$



(10)  $\sin^2 \alpha (1 - \cot^2 \alpha) = \frac{1}{\csc^2 \alpha} - \cos^2 \alpha$

$$\sin^2 \alpha - \sin^2 \alpha \cot^2 \alpha =$$

$$\sin^2 \alpha - \cancel{\sin^2 \alpha} \frac{\cos^2 \alpha}{\cancel{\sin^2 \alpha}} =$$

$$\sin^2 \alpha - \cos^2 \alpha =$$

$$\frac{1}{\csc^2 \alpha} - \cos^2 \alpha =$$





• ⑪  $\sin w (\sec w + \csc w) = \tan w + 1$

$$\sin w \sec w + \sin w \csc w =$$

$$\sin w \frac{1}{\cos w} + \cancel{\sin w} \frac{1}{\cancel{\sin w}} =$$

$$\frac{\sin w}{\cos w} + 1 =$$

$$\tan w + 1 =$$



⑫  $\cos^2 y (1 - \tan^2 y) = \frac{1}{\sec^2 y} - \sin^2 y$

$$\cos^2 y - \cos^2 y \tan^2 y =$$

$$\cos^2 y - \cancel{\cos^2 y} \frac{\sin^2 y}{\cancel{\cos^2 y}} =$$

$$\cos^2 y - \sin^2 y =$$

$$\frac{1}{\sec^2 y} - \sin^2 y =$$



## Using Fraction Decomposition

• ⑬  $\frac{\cos^2 \beta - \sin^2 \beta}{\sin \beta \cos \beta} = \cot \beta - \tan \beta$

$$\frac{\cos^2 \beta}{\sin \beta \cos \beta} - \frac{\sin^2 \beta}{\sin \beta \cos \beta} =$$

$$\frac{\cos \beta}{\sin \beta} - \frac{\sin \beta}{\cos \beta} =$$

$$\cot \beta - \tan \beta =$$



⑭  $\frac{\sin^4 x - \cos^4 x}{\sin^2 x \cos^2 x} = \tan^2 x - \cot^2 x$

$$\frac{\sin^4 x}{\sin^2 x \cos^2 x} - \frac{\cos^4 x}{\sin^2 x \cos^2 x} =$$

$$\frac{\sin^2 x}{\cos^2 x} - \frac{\cos^2 x}{\sin^2 x} =$$

$$\tan^2 x - \cot^2 x =$$





• (15)  $\frac{\csc A - \cos A}{\csc A \cos A} = \sec A - \sin A$

$$\frac{\cancel{\csc A}}{\cancel{\csc A} \cos A} - \frac{\cancel{\cos A}}{\csc A \cancel{\cos A}} =$$

$$\frac{1}{\cos A} - \frac{1}{\csc A} =$$

$$\sec A - \sin A =$$

(16)  $\frac{\cos^2 B - \sin^2 B}{\sin^2 B} = \cot^2 B - 1$

$$\frac{\cos^2 B}{\sin^2 B} - \frac{\sin^2 B}{\sin^2 B} =$$

$$\cot^2 B - 1 =$$

## Using Factorization

Review on the Three Major Types of Algebraic Factoring

Extracting the Greatest Common Factor (i.e. GCF)

- Example I:  $27x^4 + 18x^2 - 54x^3 = 9x^2(3x^2 + 2 - 6x)$
- Example II:  $8y^{10} + 16y^4 - 4y^3 = 4y^3(2y^7 + 4y - 1)$

Factoring Quadratic Expressions

- Example III:  $x^2 + 8x + 15 = (x + 3)(x + 5)$
- Example IV:  $x^2 - x - 2 = (x - 2)(x + 1)$
- Example V:  $x^2 - 9x + 20 = (x - 5)(x - 4)$
- Example VI:  $4x^2 + 21x - 18 = (4x - 3)(x + 6)$

The Difference of Two Squares

- Example VII:  $x^2 - 36 = (x - 6)(x + 6)$
- Example VIII:  $x^2 - 49 = (x + 7)(x - 7)$
- Example IX:  $9x^2 - 25 = (3x - 5)(3x + 5)$
- Example X:  $4x^2 - 81 = (2x + 9)(2x - 9)$



• ⑪  $\csc A + \csc A \cot^2 A = \csc^3 A$

$$\csc A (1 + \cot^2 A) =$$

$$\csc A \csc^2 A =$$

$$\csc^3 A =$$


⑫  $\frac{\sin^2 x + 7\sin x + 12}{\sin x + 3} = \sin x + 4$

$$\frac{(\sin x + 3)(\sin x + 4)}{\sin x + 3} =$$

$$\sin x + 4 =$$


⑬  $\frac{1 - \sec^2 \theta}{1 - \sec \theta} = 1 + \frac{1}{\cos \theta}$

$$\frac{(1 + \sec \theta)(1 - \sec \theta)}{1 - \sec \theta} =$$

$$1 + \sec \theta =$$

$$1 + \frac{1}{\cos \theta} =$$


• ⑭  $\sin x - \sin x \cos^2 x = \sin^3 x$

$$\sin x (1 - \cos^2 x) =$$

$$\sin x (\sin^2 x + \cos^2 x - \cos^2 x) =$$

$$\sin x (\sin^2 x) =$$

$$\sin^3 x =$$




$$\textcircled{21} \frac{\cos^2 x - 6\cos x + 9}{\cos x - 3} = \cos x - 3$$

$$\frac{(\cos x - 3)(\cos x - 3)}{\cos x - 3} =$$

$$\cos x - 3 =$$


$$\textcircled{22} \frac{\csc^2 \theta - 1}{\csc \theta + 1} = \frac{1}{\sin \theta} - 1$$

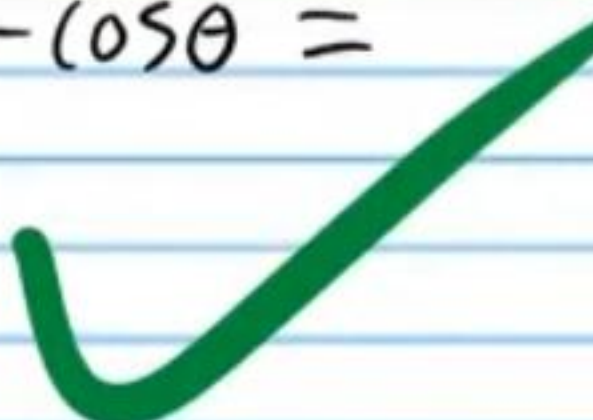
$$\frac{(\csc \theta + 1)(\csc \theta - 1)}{\csc \theta + 1} =$$

$$\csc \theta - 1 =$$

$$\frac{1}{\sin \theta} - 1 =$$


$$\bullet \textcircled{23} \frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta + \cos \theta} = \sin \theta - \cos \theta$$

$$\frac{(\sin \theta + \cos \theta)(\sin \theta - \cos \theta)}{\sin \theta + \cos \theta} =$$

$$\sin \theta - \cos \theta =$$


$$\bullet \textcircled{24} \frac{\tan^2 \theta - \cot^2 \theta}{\tan \theta - \cot \theta} = \tan \theta + \cot \theta$$

$$\frac{(\tan \theta + \cot \theta)(\tan \theta - \cot \theta)}{\tan \theta - \cot \theta} =$$

$$\tan \theta + \cot \theta =$$




# Combining Fractions

Review

$$\text{Example I)} \quad \frac{3}{8} + \frac{4}{7} = \frac{3 \cdot 7}{8 \cdot 7} + \frac{4 \cdot 8}{7 \cdot 8} = \frac{21}{56} + \frac{32}{56} = \boxed{\frac{53}{56}}$$

$$\text{Example II)} \quad \frac{1}{6} - \frac{1}{3} = \frac{1}{6} - \frac{1 \cdot 2}{3 \cdot 2} = \frac{1}{6} - \frac{2}{6} = \boxed{-\frac{1}{6}}$$

$$\text{Example III)} \quad -\frac{5}{9} + \frac{5}{6} = -\frac{5 \cdot 2}{9 \cdot 2} + \frac{5 \cdot 3}{6 \cdot 3} = -\frac{10}{18} + \frac{15}{18} = \boxed{\frac{5}{18}}$$

• (25)  $\tan \theta + \cot \theta = \csc \theta \sec \theta$

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} =$$

$$\frac{\sin \theta \cdot \sin \theta}{\cos \theta \cdot \sin \theta} + \frac{\cos \theta \cdot \cos \theta}{\sin \theta \cdot \cos \theta} =$$

$$\frac{\sin^2 \theta}{\cos \theta \sin \theta} + \frac{\cos^2 \theta}{\sin \theta \cos \theta} =$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} =$$

$$\frac{1}{\cos \theta \sin \theta} =$$

$$\frac{1}{\cos \theta \sin \theta} =$$

$$\frac{1}{\cos \theta} \cdot \frac{1}{\sin \theta} =$$

$$\sec \theta \csc \theta =$$

(26)  $\csc x - \sin x = \cot x \cos x$

$$\frac{1}{\sin x} - \frac{\sin x}{1} =$$

$$\frac{1}{\sin x} - \frac{\sin x \cdot \sin x}{1 \cdot \sin x} =$$

$$\frac{1}{\sin x} - \frac{\sin^2 x}{\sin x} =$$

$$\frac{1 - \sin^2 x}{\sin x} =$$

$$\frac{\sin^2 x + \cos^2 x - \sin^2 x}{\sin x} =$$

$$\frac{\cos^2 x}{\sin x} =$$

$$\cos x \cdot \frac{\cos x}{\sin x} =$$

$$\cos x \cot x =$$



• (27)  $\sec w - \tan w \sin w = \cos w$

$$\frac{1}{\cos x} - \frac{\sin x}{\cos x} \sin x =$$

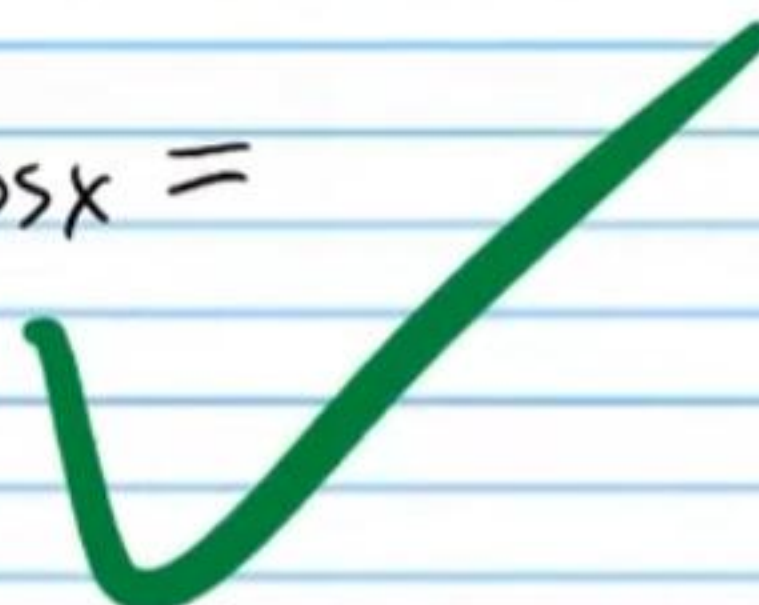
$$\frac{1}{\cos x} - \frac{\sin^2 x}{\cos x} =$$

$$\frac{1 - \sin^2 x}{\cos x} =$$

$$\frac{\sin^2 x + \cos^2 x - \sin^2 x}{\cos x} =$$

$$\frac{\cos^2 x}{\cos x} =$$

$$\cos x =$$



(28)  $\sec A - \cos A = \sin A \tan A$

$$\frac{1}{\cos A} - \frac{\cos A}{1} =$$

$$\frac{1}{\cos A} - \frac{\cos A \cdot \cos A}{1 \cdot \cos A} =$$

$$\frac{1}{\cos A} - \frac{\cos^2 A}{\cos A} =$$

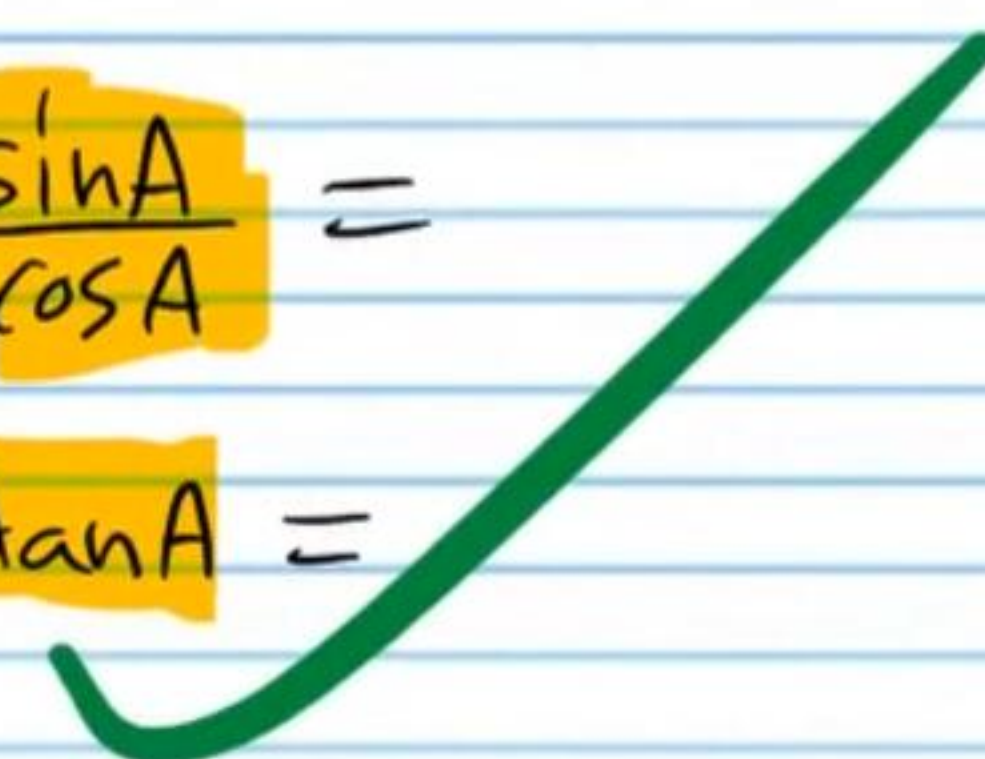
$$\frac{1 - \cos^2 A}{\cos A} =$$

$$\frac{\sin^2 A + \cos^2 A - \cos^2 A}{\cos A} =$$

$$\frac{\sin^2 A}{\cos A} =$$

$$\sin A \frac{\sin A}{\cos A} =$$

$$\sin A \tan A =$$





$$\bullet \textcircled{29} \frac{\sin x}{1 - \cos x} - \cot x = \csc x$$

$$\frac{\sin x}{1 - \cos x} - \frac{\cos x}{\sin x} =$$

$$\frac{(\sin x) \sin x}{(1 - \cos x) \sin x} - \frac{\cos x \cdot (1 - \cos x)}{\sin x (1 - \cos x)} =$$

$$\frac{\sin^2 x}{(1 - \cos x) \sin x} - \frac{\cos x \cancel{(\cos^2 x)}}{\sin x (1 - \cos x)} =$$

$$\frac{\sin^2 x - \cos x + \cos^2 x}{(1 - \cos x) \sin x} =$$

$$\frac{\sin^2 x + \cos^2 x - \cos x}{(1 - \cos x) \sin x} =$$

$$\frac{1 - \cos x}{(1 - \cos x) \sin x} =$$

$$\frac{1}{\sin x} =$$

$$\csc x =$$


$$\bullet \textcircled{30} \frac{\cos x}{1 - \sin x} - \tan x = \sec x$$

$$\frac{\cos x}{1 - \sin x} - \frac{\sin x}{\cos x} =$$

$$\frac{(\cos x) \cos x}{(1 - \sin x) \cos x} - \frac{\sin x (1 - \sin x)}{\cos x (1 - \sin x)} =$$

$$\frac{\cos^2 x}{(1 - \sin x) \cos x} - \frac{\sin x - \sin^2 x}{\cos x (1 - \sin x)} =$$

$$\frac{\cos^2 x - \sin x + \sin^2 x}{(1 - \sin x) \cos x} =$$

$$\frac{\cos^2 x + \sin^2 x - \sin x}{(1 - \sin x) \cos x} =$$

$$\frac{1 - \sin x}{(1 - \sin x) \cos x} =$$

$$\frac{1}{\cos x} =$$

$$\sec x =$$




• (31)  $\frac{1}{\cot A + 1} + \frac{1}{\tan A + 1} = 1$

$$\frac{1 \cdot (\tan A + 1)}{(\cot A + 1)(\tan A + 1)} + \frac{1 \cdot (\cot A + 1)}{(\tan A + 1)(\cot A + 1)} =$$

$$\frac{\tan A + 1}{(\cot A + 1)(\tan A + 1)} + \frac{\cot A + 1}{(\tan A + 1)(\cot A + 1)} =$$

FOIL

$$\frac{\tan A + \cot A + 2}{\cot A \tan A + \cot A + \tan A + 1} =$$

$$\frac{\tan A + \cot A + 2}{\left(\frac{1}{\tan A}\right) \tan A + \cot A + \tan A + 1} =$$

$$\frac{\tan A + \cot A + 2}{1 + \cot A + \tan A + 1} =$$

$$\frac{\tan A + \cot A + 2}{\tan A + \cot A + 2} =$$

$$1 =$$

• (32)  $\frac{1}{1 - \tan \theta} + \frac{1}{1 + \cot \theta} = \frac{2 + \cot \theta - \tan \theta}{\cot \theta - \tan \theta}$

$$\frac{(1) \cdot 1 + \cot \theta}{(1 - \tan \theta)(1 + \cot \theta)} + \frac{1 \cdot 1 - \tan \theta}{(1 + \cot \theta)(1 - \tan \theta)} =$$

$$\frac{1 + \cot \theta}{1 + \cot \theta - \tan \theta - \tan \theta \cot \theta} + \frac{1 - \tan \theta}{1 + \cot \theta - \tan \theta - \tan \theta \cot \theta} =$$

$$\frac{2 + \cot \theta - \tan \theta}{1 + \cot \theta - \tan \theta - \tan \theta \cot \theta} =$$

$$\frac{2 + \cot \theta - \tan \theta}{1 + \cot \theta - \tan \theta - 1} =$$

$$\frac{2 + \cot \theta - \tan \theta}{\cot \theta - \tan \theta} =$$